RESEARCH ROADMAP

STATUS UPDATE - 2026-10-07

How to read this update. This addendum supplements the original 22-page roadmap. It records work completed in the repository, generated evidence, and remaining dependencies. Earlier archived results remain historical records and are not silently replaced. All current outputs are simulator-derived; none is field validation or a QKD security proof.

1. Work completed in this update

Workstream	Change / result	Remarks and evidence
Corrected route comparison	Baseline selectors use the observation's feasible-action ordering; all methods use the same season and channel seeds. Per-episode output records action validity, no-feasible-action states, fallbacks, route, hops, revisits, and link validity.	The comparison output is identified below. No-feasible-action states are separated from invalid policy actions.
Physics consistency	Independent formula transcription covered 201 fiber distances from 0 to 200 km. Maximum absolute difference: QBER 0; normalized proxy 0. Transmission and proxy decrease; QBER does not decrease.	Internal consistency only. It does not validate assumptions against measurements or establish security.
FSO stochastic checks	All 16 configured season/distance anchors sampled at 2,000 trials each; maximum absolute availability deviation 0.0125 (1.22 target standard errors). Log-Cn2 AR(1): configured 0.85, sampled 0.8522 over 20,000 steps.	These are model targets and samples, not validation against Indian weather observations.
Link-feature sensitivity	The controlled diagnostic includes a proxy-rate ratio sweep, a feasible QBER-margin sweep, and an outage-mask counterfactual with route context held fixed. The FSO action probability rose from 0.311 to 0.638 as its proxy-rate ratio rose from 0.25x to 4x; at QBER 0.111 it was masked. In a separate 32-sample FSO turbulence counterfactual, a 100x Cn2 multiplier lowered median proxy from 0.202 to 0.183 and mean FSO action probability from 0.749 to 0.740; FSO remained selected in 100% of samples.	A policy-input diagnostic does not establish link-measurement validity or route-level performance.
Independent training-seed stability	Five independent checkpoints per architecture were completed. GNN mean success 46.1% (SD 16.8%; nested 95% interval 30.9%-58.5%); MLP mean success 79.8% (SD 9.4%; nested interval 71.3%-87.4%).	Variability is across fitted policies and scenario clusters in this simulator protocol; it is not external validation.
Exact toy-graph route oracle	Exhaustive simple-path reward search covered 20 graph/channel cases with a feasible oracle in 20. The policy succeeded in 90%; it exactly matched oracle reward in 55% of all cases and 61% of successful cases.	Five generated 7-node graphs; episode link states frozen with 5% per-edge outage. This exact result is limited to the toy static-channel protocol, not the full dynamic graph.
Paired endpoint / season / time transfer	Across 108 GNN episodes, success was 99/108 (91.7%; Wilson 95% CI 84.9%-95.6%); successful mean hops 31.94. BFS-hop success was 108/108 (100.0%), with 15.40 successful mean hops. Paired exact McNemar p=0.0039; among jointly successful episodes, GNN-minus-BFS hop difference was 16.00 (95% pair-cluster bootstrap CI 6.48 to 25.97). GNN did not outperform BFS.	Six ordered pairs, four seasons and selected day/night times, paired seeds. BC endpoint randomization means these are not strict end-to-end endpoint holdouts. Hop interval resamples only six endpoint-pair clusters.
Repeated inference profile	1 thread(s): p50/p95/p99 26.52/70.37/94.68 ms; 4 thread(s): p50/p95/p99 22.13/45.12/64.58 ms; 8 thread(s): p50/p95/p99 12.76/16.89/22.58 ms. Process RSS after run 348.4 MiB; model parameters 0.5 MiB.	42 ordered pairs, deterministic single-decision calls, encoder cache disabled. One host/topology; graph-size scaling and deployment profiling remain open.
Matched-budget ablation campaign	0/5 conditions complete at the shared 50-epoch budget. Current condition: baseline, ppo phase, PPO 25/50.	Single training seed; progress is not a completed comparative result.
Strict matched held-out evaluation	Five GNN and five MLP policies were evaluated on the same 108 held-out scenarios. Training-seed mean success: GNN 46.1% (SD 16.8%; nested 95% interval 30.9%-58.5%), MLP 79.8% (SD 9.4%; nested interval 71.3%-87.4%), BFS 100.0%. Mean successful hops across seeds: GNN 87.77, MLP 57.80. At 10% fiber outage, GNN/MLP mean success was 46.3%/80.6%.	Paired scenario IDs; five fitted policies per learned architecture; intervals resample training seeds outside scenario clusters. Simulator results, not field validation.

Workstream	Change / result	Remarks and evidence
Final-checkpoint reward decomposition	One final GNN seed and its matched MLP seed were decomposed over 108 scenarios against BFS-hop. Route success: GNN 46.3%, MLP 87.0%, BFS 100.0%.	Reward components are descriptive shaped-return contributions; route metrics remain the primary comparison and this one-seed audit does not estimate seed uncertainty.

2. Paired route comparison

Policy	Success	Successful mean hops	Invalid actions	No-valid-action states	Revisits
GNN-PPO	16/16 (100.0%; 95% CI 80.6%-100.0%)	28.25	0	0	2
Random	0/16 (0.0%; 95% CI 0.0%-19.4%)	n/a	0	8	2459
Dijkstra-km	4/16 (25.0%; 95% CI 10.2%-49.5%)	233.00	0	0	172
BFS-hop	16/16 (100.0%; 95% CI 80.6%-100.0%)	28.25	0	0	3
Max-SKR	0/16 (0.0%; 95% CI 0.0%-19.4%)	n/a	0	5	984

Conditions: Delhi to Chennai; 4 shared environment seeds per each of 4 seasons; QBER/rate action mask; 400 step limit. Scope is single-route simulator behavior, not traffic blocking, network-wide key-pool balance, or delivered service capacity.

Checkpoint transfer across city pairs

The same 50-epoch checkpoint completed 31/42 ordered city-pair episodes (73.8%) under the archived monsoon-night scenario. Median successful route length was 20 hops; localized decision latency was 3.83 ms median and 4.34 ms p95 on this host. This is one stochastic sample per ordered pair, not a held-out endpoint protocol or a matched baseline comparison.

3. Roadmap status after this work

Priority	Status	Evidence / remaining work
P0-1 corrected route comparison	DONE - route-level	Paired environment seeds, shared feasibility rules, and action/fallback accounting. The separate demand-driven QKP comparison remains out of scope.
P0-2 physics checks	DONE - internal only	Fiber formula consistency, configured FSO anchor sampling, and temporal correlation. No matched device/channel calibration or measured traces.
P0-3 link sensitivity	DONE - diagnostic	Rate-ratio, QBER-margin, and hard outage-mask tests vary simulator policy inputs only.
P0-4 QBER stress	PARTIAL	Controlled action-probability sweep covers QBER 0.01, 0.03, 0.05, 0.07, 0.09, 0.10, 0.109 and 0.111. The out-of-threshold candidate is masked. Route success, entropy, and preference curves across endpoint/weather conditions remain.
P0-5 FSO / outage stress	DONE - paired route stress	Paired route outcomes are archived at outage probabilities 0%, 1%, 5%, 10%. These are simulator stress scenarios; pointing/background parameters remain fixed and outage-persistence bursts are not separately calibrated.
P0-6 five training seeds	DONE - five GNN + five MLP	Five GNN and five matched MLP full-budget runs are complete and their checkpoint hashes are recorded in the strict evaluation report.
P1 endpoint / weather holdouts	DONE - strict paired test	Final report covers 108 paired held-out endpoint/condition/scenario IDs, five trained seeds each for GNN and MLP, four fiber-outage levels, and per-checkpoint CPU latency. See the final report and manuscript for estimates and uncertainty.
P1 architecture / feature ablations	DEFERRED - outside current scope	An earlier campaign stopped during the baseline at PPO epoch 25/50; no ablation comparison is used in the manuscript and the campaign is deferred.
P1 inference cost	DONE - one-host protocol	Per-decision latency was measured for all ten final checkpoints with warm-up, four CPU threads, and caching disabled. Results characterize one host/topology only; graph-size scaling and deployment profiling remain open.
P1 exact small-graph oracle	DONE - toy scope	Exhaustive simple-path reward oracle and policy regret archived for generated 7-node graphs with frozen episode channels. Does not establish optimality on the full dynamic network.
P2 network-service study	BLOCKED BY MODEL SCOPE	Requires absolute key generation and QKP capacity, time-indexed demand, consumption/refill, blocking, load balance, controller message semantics, prediction baselines, and executable demand-level ILP.
P3 external validation	BLOCKED BY DATA / HARDWARE	Requires matched device/channel measurements or hardware-in-the-loop and separate model-error analysis.

4. Interpretation and next work

The corrected comparison supports a route-level simulator statement only. Report success, successful-route hops, revisits, action-mask violations, and route outcomes; do not infer network capacity or superiority from reward alone. Preserve baseline failures as observed outcomes and state each baseline definition.

The strict five-seed endpoint/weather and paired outage evaluation, selected-checkpoint reward decomposition, final figures, and revised manuscript are complete. Before submission, the authors should fill the author/funding block and verify the final package in their target venue template. The demand/QKP service-capacity study and external device/weather validation remain deferred extensions outside this completion. Do not present the demand-level BB84/QKP formulation as implemented until an absolute key-resource and demand model is adopted and validated.

Regression check: 41 tests passed, 0 failed in 68.8 s; 1 warning(s) recorded. Details: `test_results_20261002.json`.

Provenance rule: Label every numeric result as measured, independently calculated, model-sampled, or derived from a policy run. Internal consistency checks do not upgrade an assumption to calibrated physics.